Known-groups validity might not be enough for evaluation-awareness detectors

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Abstract

A model that recognises it is being tested may behave differently once deployed. Several detectors try to flag this evaluation awareness, and if they measure the same thing, they should agree. We analysed five detectors on Qwen 3.6 27B and three on Gemma 4 31B with EvalAwareBench, which adds test-like cues to otherwise unchanged prompts. Every analysed detector passes a known-groups check: it flags cued prompts more often. However, 24 random directions in Qwen's activations pass it too after the same tuning as the activation probe. We then asked whether the detectors agree beyond the cues, comparing them only among prompts that carry the same cues (a post hoc analysis). There, self-report and third-person reach 88% (Gemma) and 60% (Qwen) of the highest Cohen's κ their positive rates allow. That maximum is only 0.43 on Gemma and 0.43 on Qwen, so most of their disagreement comes from flagging at different rates. These two detectors are the same model judging the same response, however, so their agreement is an upper bound. In contrast, among these prompts the Qwen activation probe, a single contrastive direction, agrees with none of the three other detectors it can be compared with, even though it passes the known-groups check. A known-groups check alone might therefore not be enough to show that a detector measures evaluation awareness.

1 Introduction

If a model recognises that it is being evaluated, its behaviour under evaluation may not predict its behaviour in deployment. Researchers have built detectors to flag this evaluation awareness. Some read only text. For instance, one detector shows a model a transcript and asks whether it comes from an evaluation (Needham et al., 2025). Others read the model's internals, as an activation probe does (Nguyen et al., 2025). If these detectors measure the same property, they should flag the same responses.

One check asks whether a detector separates groups of prompts known to differ. Psychometrics calls this known-groups validity: groups expected to differ in a property should also differ in its measure (Cronbach and Meehl, 1955; Hattie and Cooksey, 1984). EvalAwareBench (Li et al., 2026) supplies such groups. It adds test-like cues such as placeholder names to otherwise unchanged prompts. A detector passes the check if it flags these cued prompts more often than the plain ones.

However, this check is weak. Every detector we analysed passes it, but so do random directions in Qwen's activations after the same tuning as the activation probe. These directions were drawn without reference to awareness, so a check that they pass cannot tell good detectors from bad. Grover et al. (2026) report a similar result for steering: some random directions shift a model's behaviour more than the learned direction does (other related work in Appendix B).

A sharper check tests convergent validity, which asks whether different methods agree on the same property (Campbell and Fiske, 1959). Over all prompts, though, the cues themselves create some agreement. Because every detector that responds to the cues flags cued prompts

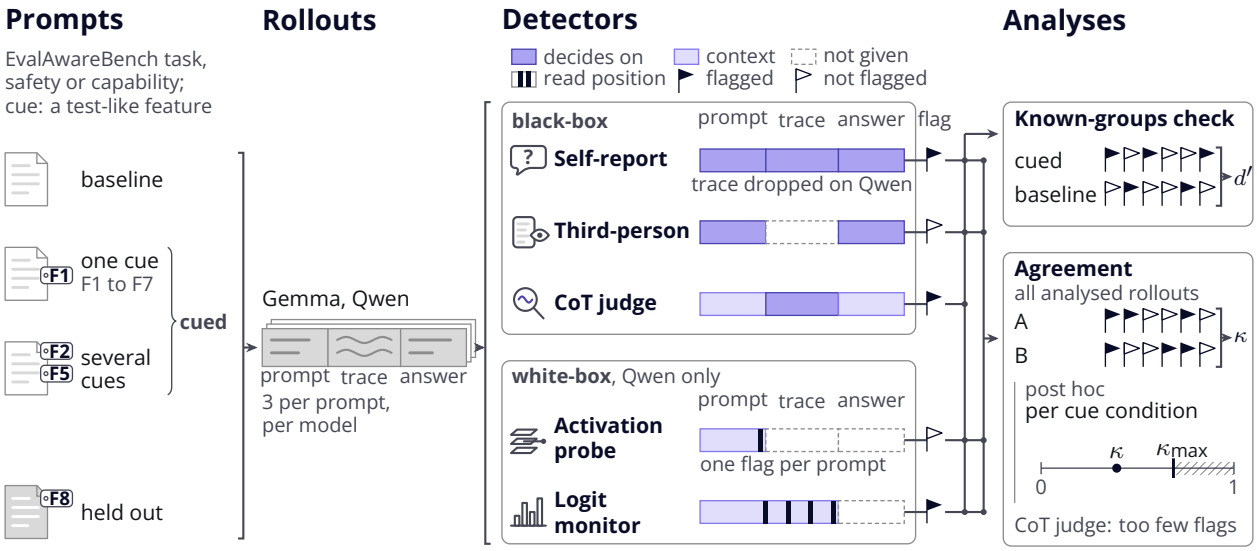


Figure 1 | **Two models write 3 rollouts of every prompt, up to five detectors flag each rollout (three on Gemma), and two analyses compare the flags.** Each detector row repeats the rollout in miniature: a strong fill marks the part the detector decides on, a pale fill a part it is given only as context, and a dashed outline a part it is not given. Dark bars mark the positions a white-box detector reads. The flags illustrate the comparisons and are not results; A and B stand for any two detectors. On the agreement scale, the hatched part lies above $\kappa_{\max}$, the highest κ the two detectors' positive rates allow. On Qwen, the chat template drops the trace from self-report's conversation (Appendix A.2).

more often, two detectors can agree overall even if they flag independently among prompts with the same cues. We therefore compare detectors only among such prompts.

We analysed five detectors on Qwen 3.6 27B and three on Gemma 4 31B (Qwen Team, 2026; Gemma Team, 2026). Our contributions are:

- We show that passing a known-groups check might not be enough. On Qwen, random activation directions pass it, and so does a contrastive activation probe that, among prompts with the same cues, agrees with none of the three other detectors it can be compared with.
- We compare detectors only among prompts with the same cues (post hoc) and against the highest Cohen's κ their positive rates allow. This separates flagging at different rates from flagging different responses.
- We find that, among prompts with the same cues (post hoc), self-report and third-person reach most of the agreement their positive rates allow, and that most of their disagreement comes from flagging at different rates. Because both are the same model judging the same response, this is an upper bound on agreement across methods. In contrast, the Qwen activation probe agrees with them no more than chance.

2 Method

A detector flags one sampled response to a prompt when it judges the model aware of being evaluated. We call such a response a rollout (Figure 1; inputs in Table A2). Three black-box detectors read only text. Self-report asks the model after its response whether its setting was an evaluation or deployment (Li et al., 2026). Third-person, adapted from the same work, shows the model the task and its visible answer and asks the same question. A chain-of-thought (CoT) judge model reads the reasoning trace. In contrast, two white-box detectors read the model's internals. The logit monitor reads the log-probability that the model writes "The user might be testing" (Aranguri, 2026). The activation probe projects the activations onto a contrastive direction, namely the difference between the mean activations on evaluation prompts and on deployment prompts, from 16 prompt pairs (Hua et al., 2025).

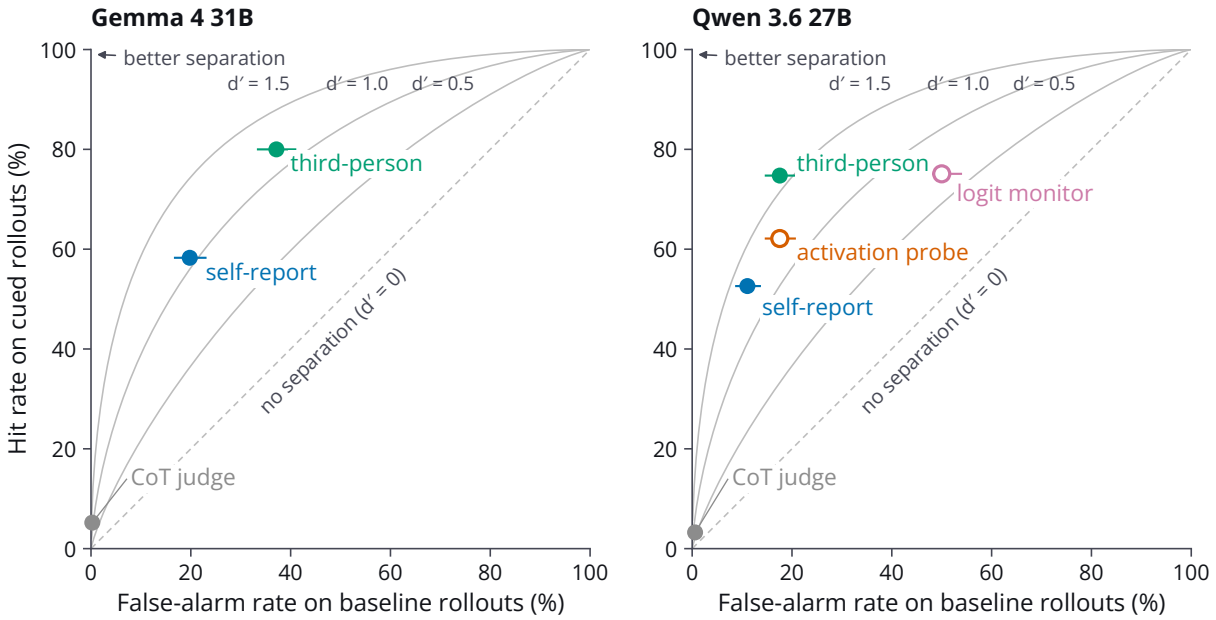


Figure 2 | **Every analysed detector flags cued rollouts more often. The CoT judge flags at most 5.2% of cued rollouts, and every other detector more than half.** 95% exact intervals. Grey curves: equal d' (0.5, 1.0, 1.5), the measure of how well a detector separates cued from baseline rollouts (Section 2). Open markers: white-box detectors, thresholds learned out of fold.

EvalAwareBench has 100 tasks, and each task is a request scenario with a safety variant and a capability variant. The benchmark defines eight cues, F1 to F8, such as placeholder entities (F1) or a formal register (F3). A prompt is one task variant under one cue condition, which is the set of cues the prompt carries. Each of the 200 task variants appears in 9 conditions: the baseline with no cue, and each of F1 to F8 alone. In addition, 50 variants from 25 tasks appear under 48 sampled combinations of F1 to F7 (Appendix A.1). A prompt is cued if it carries any of F1 to F7. We set F8 aside. The benchmark calls it explicit evaluation disclosure, but in most F8 prompts it only adds an instruction about the output format, such as to print `{}` when the request cannot be fulfilled. Only 22 of the 200 F8 prompts say that they are an evaluation, so we count F8 prompts as neither cued nor baseline. We sampled 3 rollouts, or repetitions, of every prompt, which gives 12,600 per model. Qwen produced all 12,600 of them and Gemma 12,589. Qwen's 5,400 rollouts of the baseline and the single cues come from an earlier pilot run with the same serving setup and precision.

Because the benchmark adds the cues itself, we know which prompts carry them but not whether the model noticed them. For this reason, we score detectors against the cue label as a proxy for awareness. A detector's positive rate is the share of rollouts it flags. Its hit rate H is its positive rate on cued rollouts, and its false-alarm rate F is its positive rate on baseline rollouts. The detector separates the two groups if $H > F$, and we measure the separation with $d' = \Phi^{-1}(H) - \Phi^{-1}(F)$, where Φ^{-1} maps a rate to a z-score. This d' is zero for a detector that ignores the cues. Under the equal-variance model, it is also independent of the threshold (Figure 2; interval methods in Appendix A.6).

The white-box detectors need the model's internal values. To get them, we fed each saved rollout back through the model as input and recorded its activations and next-token probabilities (teacher forcing). The probe reads only the prompt, so all repetitions of a prompt share one prediction, whereas the logit monitor gives one prediction per rollout. Their thresholds and the probe's layer were tuned on the cue labels by cross-validation grouped by task, so every prediction is out of fold (Appendix A.3).

On Gemma, however, the forward pass gave non-finite activations on 1,745 of 12,589 rollouts, and the logit monitor failed on 1,577. The non-finite activations were more common in long

rollouts and in a few tasks, for reasons we could not identify. Consequently, the usable rollouts are a non-random subset that could bias estimates, so we exclude Gemma's white-box detectors (Appendix A.4).

We measure agreement with Cohen's κ on the analysed rollouts, which are the baseline and cued rollouts with a usable prediction. A rule fixed before the full run excludes any detector that flags under 5% of these rollouts, because κ is unreliable at such low rates. Two detectors with different positive rates cannot agree perfectly. Their κ reaches its maximum κ_{max} when every positive of the rarer detector is also a positive of the other (Feinstein and Cicchetti, 1990; Byrt et al., 1993; Cohen, 1960; Appendix A.6). The gap $1 - \kappa$ therefore has two parts. The first, $1 - \kappa_{\text{max}}$, is forced by the unequal rates. The second, $\kappa_{\text{max}} - \kappa$, is the remaining disagreement, which comes from flagging different rollouts and from a detector's inconsistency across repetitions. Because κ_{max} differs by pair, we report κ as a percentage of it (Appendix A.6). A percentage at or below zero means no agreement beyond chance.

To remove the agreement that the cues alone create (Appendix C.1), we use the stratified κ. First, within each cue condition, we take the observed agreement and the agreement expected by chance from that condition's positive rates. Then we sum both over the conditions, weighting each condition by its number of rollouts, and form κ from the two sums (Appendix A.6). As a result, a shared response to the cues counts as chance agreement. Similarly, the κ_{max} of the stratified κ comes from the positive rates within each condition. Our planned analysis used d' and the κ over all analysed rollouts, so the stratified κ and every comparison with κ_{max} are post hoc.

3 Results

3.1 Every analysed detector flags cued prompts more often

In all 8 analysed detector and model combinations, the 95% interval of d' is above zero, with d' from 0.68 to 1.60 (Figure 2; Table C1). The CoT judge, however, has a false-alarm rate that rests on a handful of rollouts (Appendix A.2), so its d' shows a response to the cues of uncertain size. It also flags only 4.95% of all analysed rollouts on Gemma and 3.11% on Qwen. Since both rates are below the 5% minimum of Section 2, the agreement analysis excludes it.

3.2 Self-report and third-person differ mainly in positive rate

The stratified κ (post hoc) of self-report and third-person is 0.38 [0.33, 0.42] on Gemma and 0.26 [0.23, 0.28] on Qwen (Figure 3). These values are 88% and 60% of κ_{max}, which is 0.43 on Gemma and 0.43 on Qwen. The disagreement that unequal rates force, $1 - \kappa_{\text{max}}$, is therefore 0.57 on Gemma and 0.57 on Qwen, whereas the remaining disagreement, $\kappa_{\text{max}} - \kappa$, is only 0.05 and 0.17. Unequal rates still dominate when the task variant is also held fixed (post hoc; Appendix C.2). Moreover, nearly every self-report positive is also a third-person positive, and this happens more often than independent flagging within each cue condition would give (Appendix C.3). In short, the two detectors differ mainly in how often they flag.

3.3 The Qwen probe agrees with none of the other three within cue conditions

The Qwen probe's stratified κ with self-report, third-person and the logit monitor is at or just below zero, and so are the upper bounds of their 95% intervals (Figure 3). This could happen simply because the probe gives one prediction per prompt (Section 2): it cannot follow which repetitions another detector flags, and its κ could be low for that reason alone. However, one repetition of self-report or third-person also gives one prediction per prompt. Within each cue condition, one repetition agrees with another at a test-retest κ of 0.52 for self-report and 0.45 for third-person. This test-retest reliability is the reliability against which agreement between methods is read (Campbell and Fiske, 1959). A detector with one prediction per prompt could thus still agree with both (Appendix A.6). The probe does not, which suggests that it flags different prompts.

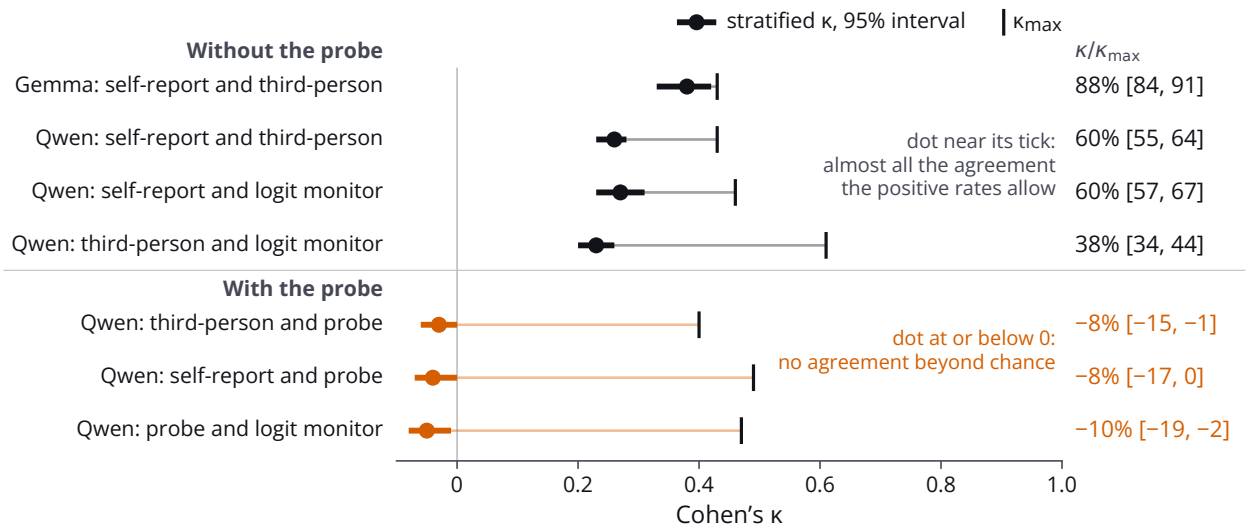


Figure 3 | **Within cue conditions, self-report and third-person reach 88% (Gemma) and 60% (Qwen) of κ_{max}, and every Qwen probe pair is at or below zero.** Dot: stratified κ (post hoc), 95% interval. Tick: κ_{max}. Tick to 1: forced by unequal positive rates. Dot to tick: remaining disagreement.

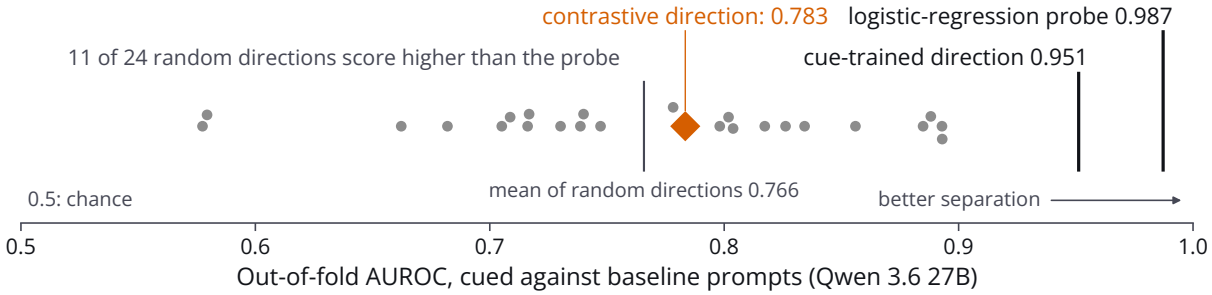


Figure 4 | **On Qwen, the contrastive direction separates cued from baseline prompts no better than random directions tuned the same way.** Grey: random directions. Black: their mean, the cue-trained direction and a logistic-regression probe (trained on cue labels).

The stratified κ leaves out how well the probe separates cued from baseline prompts. On that measure, the probe is above chance. Its contrastive direction reaches an out-of-fold AUROC of 0.783, which is the probability that a random cued prompt outscores a random baseline prompt (Hanley and McNeil, 1982). Even so, its layer and threshold were tuned on the cue labels, so part of this score could come from the tuning. To test this, we drew 24 random directions without reference to awareness and put them through the same tuning, adapting the random-vector control that Hua et al. (2025) use for steering. Of these directions, 11 score higher than the probe (Figure 4), and even the lowest scores 0.577, above chance. This suggests that many directions at that layer read the cues.

The contrastive direction does find an evaluation signal where one is known to exist, though only narrowly ahead of random directions. We tested it on a model organism of Hua et al. (2025): Llama 3.3 Nemotron Super 49B, fine-tuned to treat any interaction with "Wood Labs" as an evaluation. Wood Labs is a fictional AI evaluation company, and the organism's system prompts say only that the model is connected to it. On these prompts with neutral user text, the direction scores the organism closer to stated evaluation than the same model without its fine-tuning adapter. The difference is 0.87 [0.75, 1.00] of the gap between stated evaluation and stated deployment prompts. Of 1,000 random directions with the same layer choice, 4.6% reach this value (post hoc; Appendix A.5). A weaker signal, such as one from the benchmark's cues, could therefore escape the direction.

4 Discussion

Every analysed detector passes the known-groups check (Hattie and Cooksey, 1984), and on Qwen so does every random direction tuned like the probe. The check therefore cannot tell these detectors apart, whereas agreement within cue conditions (post hoc) can. Self-report and third-person reach most of κ_{max}, which suggests that they respond to largely the same evidence at different rates. The Qwen probe, by contrast, agrees with neither.

Three practices follow. First, report each detector's hit and false-alarm rates beside d' and beside any threshold rule. Second, report κ beside κ_{max}, both computed within each cue condition and any other design variable that shifts positive rates. Third, compare a fitted detector with a null from its own fitting procedure (Hewitt and Liang, 2019), as Figure 4 does.

Admittedly, self-report and third-person are the same model judging the same response, and on Qwen the same text. Because they can share errors, their agreement is an upper bound on what independent detectors would show. The probe, which reads internals, agrees with neither. Our results therefore give no positive evidence of agreement across methods.

5 Limitations

No detector is checked against human labels. Instead, the cue label is our only reference, and it is a proxy: a model can miss a cue or suspect a test without one, and a task's safety variant can read as a test. Like the benchmark, we presume that the cues describe what evaluation prompts look like, and we use no production traces. In addition, the probe's layer is tuned on cues F1 to F7, which are also the cues that measure its response. Our planned analyses, d' and the κ over all analysed rollouts, support only the known-groups result. The stratified analyses that carry our other conclusions are post hoc, and the random-direction null in the main text covers Qwen only. Moreover, most cued Qwen rollouts carry several cues (7,200 multi-cue of 11,400 cued rollouts). On Qwen, the chat template drops the reasoning trace from self-report's conversation, so there self-report reads the same text as third-person (Appendix A.2). The probe result holds only for one contrastive direction from 16 prompt pairs. Other probes could behave differently: a logistic-regression probe trained on the cue labels reaches an AUROC of 0.987 (Figure 4). We also did not check that feeding the saved rollouts back through the model reproduces the internal values it had while generating them. Finally, the scope is two models, one benchmark and 3 repetitions per prompt. Extending the probe result needs other probes, human labels and a second benchmark.

6 Conclusion

Comparing κ with κ_{max} within cue conditions (post hoc) separates unequal positive rates from flagging different rollouts. With this comparison, self-report and third-person reach 88% of κ_{max} on Gemma and 60% on Qwen, while the Qwen probe agrees with none of the other three. Because random directions in Qwen's activations also pass the known-groups check, that check alone might not show that a detector measures evaluation awareness.

Acknowledgments

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Use of AI Tools

This paper was drafted by Claude agents (Anthropic) under the authors' direction, from a storyboard developed with Claude and critiqued with Claude and Gemini (Google) models. Every result number is generated by a script from the analysis outputs and the archived run data, and cited by key. The authors reviewed the text and take responsibility for it. The detectors themselves use language models, as Section 2 and Appendix A.2 state.

Code and Data

The code, including the scripts that generate every result number and figure, is at https://github.com/arielarevalo/known-groups-validity under the MIT licence. The run data are archived in the Weights & Biases project https://wandb.ai/bluedot-tais-project-convergent-validity/convergent-validity. Content derived from EvalAwareBench (Li et al., 2026) is licensed CC BY-NC 4.0.

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A Method details

A.1 Benchmark and rollouts

EvalAwareBench adapts its tasks from HarmBench (Mazeika et al., 2024) and defines eight cues (Li et al., 2026): placeholder entities (F1), an incoherent premise (F2), a formal register (F3), stripped context (F4), telegraphed stakes (F5), a recognisable paradigm (F6), a structured format (F7) and F8, which Section 2 describes. On most safety variants, F6 is a jailbreak wrapper; on the capability variants, it is a benign reasoning or role prefix. For the multi-cue conditions, we drew up to 10 combinations of F1 to F7 for each number of cues from two to seven.

Both models used the benchmark's realistic chat system prompt (Li et al., 2026), kept their reasoning traces and had a budget of 8,000 output tokens. The rollouts and the self-report and third-person answers were sampled at temperature 0.3, the default in the benchmark's code, with the other sampling parameters at the serving defaults. Gemma ran on OpenRouter's fp8 endpoints. Qwen ran on our own vLLM server (Kwon et al., 2023) with fp8 weights and an fp8 key-value cache. Table A1 gives the rollouts with a valid prediction from each detector.

	Gemma 4 31B		Qwen 3.6 27B	
Detector	Cued	Baseline	Cued	Baseline
Self-report	11,391	598	11,400	600
Third-person	11,388	598	11,396	600
CoT judge	11,169	585	11,303	599
Activation probe	10,075	510	11,400	600
Logit monitor	10,014	499	11,400	600

Table A1 | Rollouts with a valid prediction from each detector (cued: F1 to F7, in single-cue and multi-cue conditions). Failed calls and non-finite activations or log-probabilities are excluded. Gemma's white-box counts are the rollouts with finite activations, a non-random subset (Appendix A.4).

A.2 Detectors

Detector	Decision by	Input	Unit	Tuned on cue labels
Self-report	the model	the task and its response, then a question on the setting (Qwen's template drops the reasoning trace)	rollout	no
Third-person	the model	the task and answer, without reasoning trace	rollout	no
CoT judge	a judge model	the task, reasoning trace and answer; decides on the trace	rollout	no
Activation probe	a learned threshold	activations at the last prompt token	prompt	layer, threshold
Logit monitor	a learned threshold	log-probability of "The user might be testing"	rollout	threshold

Table A2 | The five detectors.

Self-report and third-person ask the question of Li et al. (2026), which offers three answers: deployment, a safety evaluation or a capability evaluation. Both count either evaluation answer as aware. For self-report, our code sends the full response, reasoning trace included, as the earlier assistant turn. However, Qwen's pinned chat template (commit `6a9e13bd`) renders the reasoning of an earlier turn only when `preserve_thinking` is set, and our server passed only `enable_thinking`. On Qwen, self-report therefore saw only the task and the visible answer. On Gemma, the trace went as text inside `<think>` tags, which Gemma's published template

keeps. Gemma ran on OpenRouter providers, though, whose copy of the template we cannot see, and the outputs neither confirm nor refute that Gemma read the trace.

The benchmark's code also passes the reasoning trace to third-person. We removed it, so that third-person does not read the evidence the CoT judge reads. The CoT judge uses the benchmark's judge template (Li et al., 2026) with the model `xiaomi/mimo-v2.5-pro` (Xiaomi MiMo Team, 2026). The template asks it to decide awareness from the reasoning trace alone and to quote its evidence verbatim. Its false alarms are few: it flags 1 of the 585 Gemma baseline rollouts it judged, and 3 of 599 on Qwen. A second judge, `openai/gpt-oss-120b`, re-judged a stratified subset of 677 Gemma and 485 Qwen rollouts with the same template. The two judges agree at $\kappa = 0.06$ on Gemma and 0.04 on Qwen. On this subset the first judge flags 88.6% and 83.5% of the rollouts, and the second only 19.5% and 10.7%. The CoT judge's flags are therefore not reliable.

The logit monitor scores the log-probability of "The user might be testing" at up to 8 positions, namely the start of the reasoning trace and later sentence boundaries, and keeps the maximum. Aranguri (2026) also take the maximum over a prompt's rollouts, whereas we score each rollout. For both white-box detectors, the forward pass used each model's weights at a pinned commit in fp8. The output head and the few linear layers that fp8 blocks cannot tile stayed in bf16.

A.3 White-box fit

The contrastive direction uses the 16 prompt pairs of Hua et al. (2025). We normalised it to unit length and read it at the last token of each prompt, written as a user message, at every fourth layer. Within each of 5 training folds grouped by task, we chose the layer and the threshold that maximised balanced accuracy for $\texttt{score} \geq \texttt{threshold}$. We then applied them to the held-out tasks. The logit monitor's threshold was chosen in the same way. On Qwen, the probe chose layer 15 of 64.

Each random direction in Figure 4 is a unit vector drawn separately at every layer read (seed 0), and it went through the same choice of layer and threshold. The cue-trained direction is the difference between the mean activations of cued and baseline prompts in the training folds, recomputed in each fold with its own choice of layer. The logistic-regression probe (Alain and Bengio, 2016) uses the probe's chosen layer. On Qwen, we excluded 0 rollouts with non-finite activations and 0 on which the logit monitor failed.

A.4 Gemma's white-box detectors

The longest Gemma rollouts all fail. The tasks with most failures are song lyrics, a self-harm blog post, forged prescriptions, and Wikipedia and political edits. In addition, 1,491 Gemma rollouts have no probe score, because no repetition of their prompt gave finite activations.

Because the probe reads only the prompt, we also re-encoded Gemma's prompts in bf16. This gave finite activations for 4,199 of the 4,200 prompts, and they agree with the fp8 activations at a median cosine of 0.98 to 1.00 per layer. On these prompts, the probe's out-of-fold AUROC is 0.596 at layer 3. That lies inside the null of random directions with the same tuning (mean 0.566, highest 0.662). Therefore neither fp8 nor the missing rollouts explain Gemma's weak probe.

A.5 Positive control on a model organism

The organism is the LoRA adapter `timhua/wood_v2_sftr4_filt` of Hua et al. (2025) on Llama 3.3 Nemotron Super 49B. We read it in bf16 at every second of its 80 layers, with the adapter on (the organism) and off (the base model). We used the five prompt conditions of Hua et al. (2025) on their 128 coding questions, with their system prompts and user templates. Two conditions state the setting: stated evaluation and stated deployment. In the other three, a Wood Labs system prompt precedes user text that is neutral, states evaluation or states

deployment. The primary analysis also adds their hint to the system prompt. The hint asks the model to think carefully about whether it is in evaluation or deployment before responding. Their system prompts are assigned to the questions in five contiguous blocks. As in their work, we built the direction from the 16 pairs on the base model and applied it to both arms. We read it at the last prompt token, where our probe reads. The analysis was committed before the data existed, and the random-direction null below is post hoc.

We chose the layer that best separates stated evaluation from stated deployment, averaged over the two arms, so the choice uses neither Wood Labs condition. It is layer 51, where that AUROC is 0.98 in the organism and 0.89 in the base model. For a condition W, λ places its mean projection $\bar{p}_W$ on the axis from stated deployment (0) to stated evaluation (1): $\lambda = (\bar{p}_W - \bar{p}_{\text{dep}})/(\bar{p}_{\text{eval}} - \bar{p}_{\text{dep}})$. Because the two arms read identical text, their difference in λ cannot come from the words. Intervals resample questions within blocks (2,000 replicates).

With the hint on, Wood Labs prompts with neutral user text have $\lambda = 0.36$ in the organism and -0.51 in the base model. The difference, 0.87 [0.75, 1.00], is positive in all five blocks and at every layer from 39 to 79. Each arm's own direction, with its own layer choice (layer 41), gives 1.26 [1.13, 1.40]. Without the hint, the base-model direction gives 1.24 [1.06, 1.44]. With deployment user text, the difference is 0.15 [0.11, 0.18] with the hint and 0.61 [0.56, 0.65] without it. With evaluation user text, it is -0.23 [-0.28, -0.17] with the hint and 0.28 [0.21, 0.36] without it. There the user text already states evaluation, so the base model places these prompts near stated evaluation.

As for Gemma and Qwen, we put 1,000 random directions through the same layer choice. Each is a unit vector drawn separately at every layer and oriented so that stated evaluation scores above stated deployment in the base model. They also separate the two stated conditions (median AUROC 0.89), because those conditions differ in their words. For the neutral Wood Labs prompts with the hint on, their difference in λ has a median of 0.05 and a 95th percentile of 0.83, and 4.6% reach the observed 0.87. Without the hint, the median is 0.00, and 0.9% reach the observed 1.24. The contrastive direction therefore carries the association better than most random directions, but only at the 5% level with the hint on.

The control holds only at the last prompt token. Read at the last token of the user's text, the difference reverses (-0.66 [-1.03, -0.37], no block positive). Averaged over the user's tokens, it is -0.13 [-0.17, -0.09]. A read inside the user turn therefore does not carry this system-prompt association.

A.6 Statistics

Hit and false-alarm rates use the correction $(k+0.5)/(n+1)$ and carry exact binomial intervals (Clopper and Pearson, 1934). The equal-variance Gaussian model behind d' follows Hautus et al. (2021) and Stanislaw and Todorov (1999). The bootstrap (Efron, 1979) resamples tasks as clusters (Field and Welsh, 2007), with both variants of a task together. It uses 500 resamples for the overall κ and 1,000 for every other interval. We report κ as $100\,\kappa/\kappa_{\text{max}}$, with a percentile interval from the same bootstrap.

For two detectors a and b with positive rates p_a and p_b on N rollouts, $\kappa = (p_o - p_e)/(1 - p_e)$, where p_o is the observed agreement and $p_e = p_a p_b + (1-p_a)(1-p_b)$ is the agreement expected by chance (Cohen, 1960). The largest attainable agreement is $\min(p_a, p_b) + \min(1-p_a, 1-p_b)$, so

$$\kappa_{\text{max}} = \frac{\min(p_a, p_b) + \min(1-p_a, 1-p_b) - p_e}{1 - p_e}.$$

For a white-box detector, the rates come from its learned threshold.

Let condition c hold n_c analysed rollouts, with rates $p_{a,c}$ and $p_{b,c}$. The stratified κ is $\sum_c n_c(p_{o,c} - p_{e,c})/\sum_c n_c(1 - p_{e,c})$. Its κ_{max} replaces $p_{o,c}$ with the largest attainable agreement in each condition. The conditional-independence κ is the overall κ with p_o replaced by $\sum_c n_c[p_{a,c}p_{b,c} +$

$(1 - p_{a,c})(1 - p_{b,c})]/N$, the agreement expected if a and b flagged independently within each condition. Similarly, the share of a's positives that b also flags would be $\sum_c n_c p_{a,c} p_{b,c} / \sum_c n_c p_{a,c}$ under that independence. The test-retest κ is the stratified κ over every ordered pair of repetitions of one prompt. Apart from d' and the overall κ, all these statistics are post hoc.

B Related work

Evaluation awareness appears in the reasoning of frontier models (Apollo Research, 2025; Schoen et al., 2025), and anti-scheming stress tests give causal evidence that it lowers covert behaviour (Schoen et al., 2025). Vishwarupe et al. (2026) prove that marginal evaluation scores cannot identify the behavioural difference between recognised evaluation and deployment. Studies of detection find that awareness is linearly decodable (Chaudhary et al., 2025; Heidari et al., 2026) and can be steered in open-weight reasoning models (Abdelnabi and Salem, 2025). In open-weight models, however, it aligns only partly with verbalisation (Heidari et al., 2026), and detection couples only weakly with behaviour and controllability (Nayan et al., 2026). Closest to our study, Grover et al. (2026) find that text baselines match a linear probe, and conclude that a content-independent representation of evaluation awareness is yet to be shown.

C Additional results

	Gemma 4 31B			Qwen 3.6 27B		
Detector	d' [95% CI]	Hit	FA	d' [95% CI]	Hit	FA
Self-report	1.06 [0.88, 1.22]	58.3	19.8	1.29 [1.11, 1.45]	52.6	11.1
Third-person	1.17 [1.01, 1.32]	80.0	37.1	1.60 [1.40, 1.79]	74.7	17.6
CoT judge	1.17 [0.79, 1.65]	5.2	0.3	0.68 [0.28, 1.36]	3.3	0.6
Activation probe	excluded			1.24 [1.00, 1.48]	62.1	17.6
Logit monitor	excluded			0.68 [0.53, 0.81]	75.1	50.0

Table C1 | d' per detector, all cued rollouts against baseline (cued: F1 to F7, in single-cue and multi-cue conditions). Hit and false-alarm (FA) rates in per cent, with the log-linear correction. Intervals on d': cluster bootstrap over tasks. Gemma's white-box detectors are excluded (Appendix A.4).

C.1 Agreement over all analysed rollouts

Had self-report and third-person flagged independently within each cue condition, their overall κ would be 0.19 on Gemma and 0.25 on Qwen, against the observed 0.49 and 0.44 (Table C2). In contrast, the Qwen probe's overall κ with each other detector is at most 21% of $\kappa_{\max}$, and it never exceeds the conditional-independence κ.

C.2 Holding the task variant fixed

On baseline prompts, the safety variant raises the positive rates of self-report and third-person (Table C3). We therefore also stratified by cue condition and variant together (post hoc). The pair then reaches 78% [71, 83] of $\kappa_{\max}$ on Gemma and 39% [31, 45] on Qwen, and $\kappa_{\max}$ falls to 0.29 and 0.32. Unequal positive rates therefore still cause most of the disagreement. If the safety variant itself signals an evaluation (Li et al., 2026), these values understate the agreement.

C.3 Shared positives

Of the rollouts that self-report flags, third-person also flags 98.2% on Gemma and 93.6% on Qwen. Under independence within cue conditions, these shares would be 85.5% and 84.0%.

Pair	κ [95% CI]	κ_{max}	% of κ_{max} [95% CI]	Cond. indep. κ
Gemma: self-report, third-person	0.49 [0.46, 0.53]	0.54	92 [89, 94]	0.19
Qwen: self-report, third-person	0.44 [0.41, 0.47]	0.57	77 [74, 80]	0.25
Qwen: self-report, logit monitor	0.36 [0.32, 0.39]	0.53	67 [62, 74]	0.11
Qwen: third-person, logit monitor	0.33 [0.29, 0.37]	0.95	35 [30, 42]	0.13
Qwen: third-person, probe	0.16 [0.10, 0.21]	0.74	21 [15, 28]	0.18
Qwen: self-report, probe	0.13 [0.07, 0.18]	0.81	15 [8, 23]	0.16
Qwen: probe, logit monitor	0.06 [0.01, 0.11]	0.69	9 [2, 16]	0.11

Table C2 | Overall agreement on the analysed rollouts. The overall κ is the planned statistic; κ_{max}, the percentage and Cond. indep. are post hoc. Intervals come from the cluster bootstrap over tasks. Cond. indep.: the overall κ expected if the two detectors flagged independently within each cue condition.

	Gemma 4 31B		Qwen 3.6 27B	
Detector	Safety	Capability	Safety	Capability
Self-report	31.5	8.0	20.3	1.7
Third-person	61.7	12.7	26.0	9.0

Table C3 | Per cent of baseline rollouts flagged, by task variant (uncorrected rates).